Diseases of Gladiolus

1. **Wilt/yellow/brown rot /dry rot - *Fusarium oxysporum* f. sp. *gladioli***
2. **Corm rot - Several fungi**

   Core or Spongy Rot - *Botrytis gladiolorum*

   Dry or Neck Rot - *Stromatinia gladioli*

   Storage Rot - *Penicillium gladioli*

Diseases of Asters

1. **Anthracnose - *Colletotrichum gloeosporioides***
2. **Phyllody - Phytoplasma**
3. **Aster yellows - Phytoplasma**

Wilt/Fusarium yellows/brown rot /dry rot - *Fusarium oxysporum* f. sp. *gladioli*

Figure 1. Gladiolus plants grown from corms infected with Fusarium rot. Infected leaves are commonly weak, stunted, turn yellow, and die back starting at the tips (J.L. Forsberg photo).

Wilt/Fusarium yellows/brown rot /dry rot - *Fusarium oxysporum* f. sp. *gladioli*

Wilt/Fusarium yellows/brown rot /dry rot - *Fusarium oxysporum* f. sp. *gladioli*

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Wilt/Fusarium yellows/brown rot /dry rot - *Fusarium oxysporum* f. sp. *gladioli*

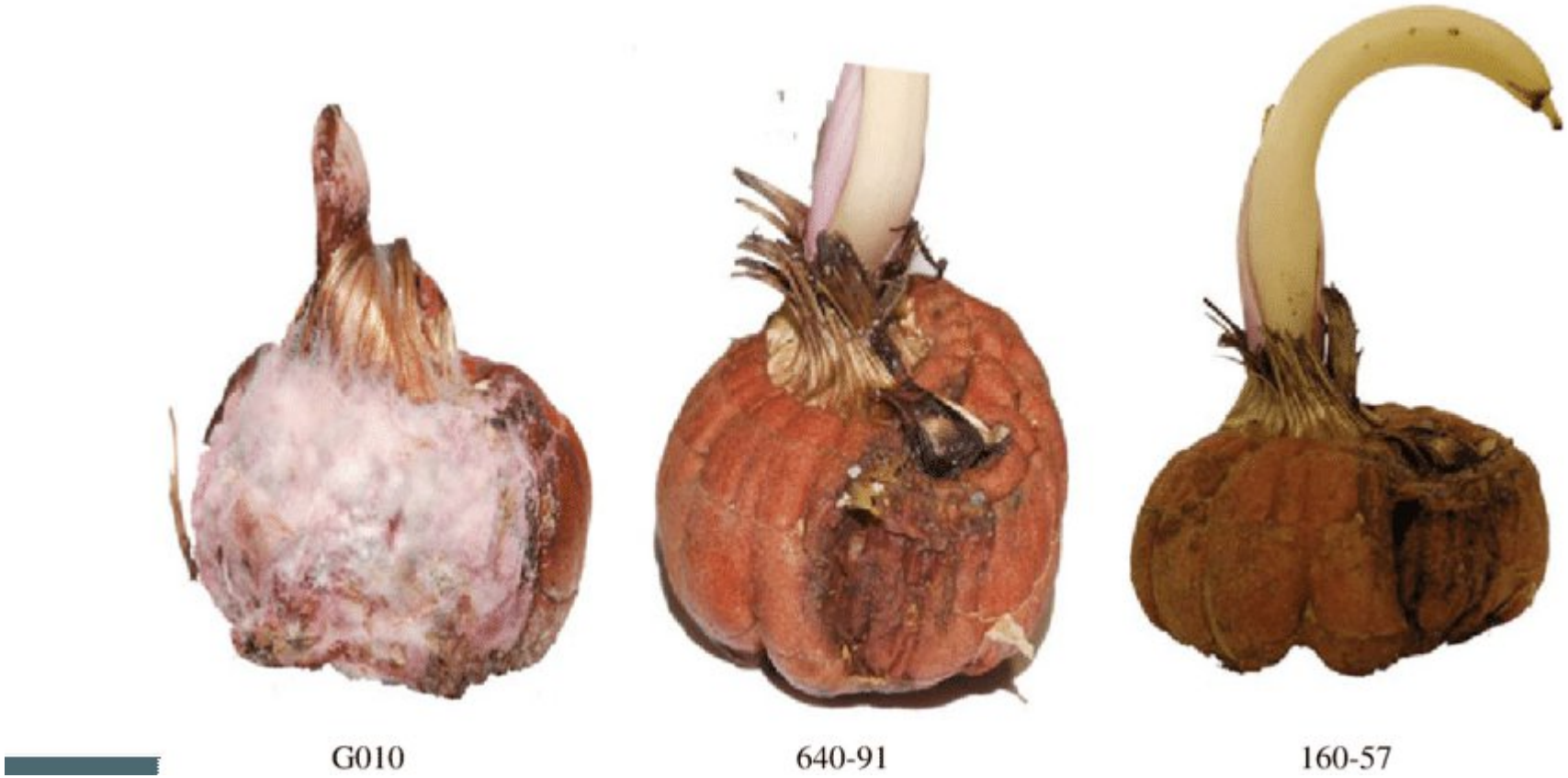
Cross section of older corms showing rotted cores and discoloured vascular streaks

Figure 2B. Brown rot form of the disease. The rotted tissue is quite thick.

Wilt/Fusarium yellows/brown rot /dry rot - *Fusarium oxysporum* f. sp. *gladioli*

Figure 2C: Basal dry-rot form - Sectioned 'spotlight' corms show the extreme thinness of the rotted tissue in contrast to the brown-rot phase.

Wilt/Fusarium yellows/brown rot /dry rot - *Fusarium oxysporum* f. sp. *gladioli*



Galdiolus-corms-30-days-after-inoculation *with F. oxysporum* f sp

Wilt/Fusarium yellows/brown rot /dry rot - *Fusarium oxysporum* f. sp. *gladioli*

Symptoms

- It causes curving, bending, arching, stunting, yellowing or drying of leaves associated with root and corm rot in field as well as in storage.
- The disease is characterized by **dry rot in storage, often restricted to the corm base.**
- Corms when split into halves, **show radiating dark- colored streaks** extending from the corm base through the flesh
- In severe cases **the centre of entire corm is black and rotten.**

Wilt/Fusarium yellows/brown rot /dry rot - *Fusarium oxysporum* f. sp. *gladioli*

Symptoms contd…

On leaves

- **Due to rotting of the under-ground parts**, yellowing of leaves and distortion or modification of the leaves, stem and flowers
- **The plant may also have blackened areas at the base** which spread onto the corm
- **Plants bend geotropically** with **leaf tip burn** is a conspicuous feature.
- The bending always starts from the site of the **corms showing rot.**
- **Yellowing of older/ outside** and the apparent **stunting of newer leaves**.
- In older plants **bending of young leaf stalks, cupping of leaf stalks** and are **often crooked** and **greener just below the florets**.
- In adverse weather conditions **bright yellow interveinal streaks on leaves**.

Wilt/Fusarium yellows/brown rot /dry rot - *Fusarium oxysporum* f. sp. *gladioli*

Symptoms contd…

On flowers:

- Infected plants either **failed to produce flowers** or **very small spikes with un opened flowers or spikes with few, small, malformed flowers**
- In addition, the **spike itself will often be stunted and faded in colour.**
- In case of **slight corm infection** the floret shape, size and colour get modified

On corms

- Lesions on the corm are **brown with well defined margin and shriveled**.
- In partially infected plants, **size of the corm was reduced** with black discoloration
- In severe infection, rotting of entire corm in the soil.

Wilt/Fusarium yellows/brown rot /dry rot - *Fusarium oxysporum* f. sp. *gladioli*

Disease cycle

PSI:

- The initial infection of the **corms comes either through soil or by latent corm** infection of the previous year.
- Fungus is carried practically in all **stocks of corms and cormels** as latent infection

SSI :

- The pathogen is also disseminated through **soil, contaminated water**

Wilt/Fusarium yellows/brown rot /dry rot - *Fusarium oxysporum* f. sp. *gladio*

Management

- **Water soaking of cormels at 57.2^0** C up to 30 minutes reduces the disease in corms lifted in mid April, or early summer and treated 6 to 10 weeks after harvest.
- **Pre planting or post harvest** dusting of corms with 10 to 20 per cent TBZ (Thiabendazole) is very effective.
- Disease can be effectively controlled by **dipping of corms in 0.1 per cent Carbendazim before planting.**
- Plant extract of *Allium sativum* and *Ocimum sanctum* has provided 61-67 % disease control

Wilt/Fusarium yellows/brown rot /dry rot - *Fusarium oxysporum* f. sp. *gladioli*

Management contd…

- Exposing the soil to the sun during summer months by deep ploughing
- Soil disinfestation/**flooding the field up to 30 cm for 3 to 4 months**
- Destruction of diseased plant parts
- Proper drainage should be maintained in the field.
- Care should be taken to prevent the movement of water from infected to healthy plants
- Application of **phosphate fertilizers and nitrate fertilizers** help in reducing wilt incidence whereas **ammonium fertilizers increase disease development**

Wilt/Fusarium yellows/brown rot /dry rot - *Fusarium oxysporum* f. sp. *gladioli*

- Amend acid soils (pH < 6.5) - agricultural limestone @1- 2 t/ac raising of soil pH to 8.2
- Soil solarization (25 to 50 µm)
- Application of organic amendments like castor cake, mustard cake and poultry manure before sowing @ 2.5 t/ha

Biological methods

- Seed treatment with *P. fluorescens* at 10 g/kg, *T. viride* (4 g/kg) and soil application (5 kg/ha)
- Pre planting treatment of corms with non-pathogenic *Fusarium* (Fo 52) and soil application.
- The combination of Fo 52 and *P. fluorescens*.

Wilt/Fusarium yellows/brown rot /dry rot - *Fusarium oxysporum* f. sp. *gladioli*

- PGPR- *Serratia marcescens, Pseudomonas putida, Penicillium oxalicum* and *Trichoderma* spp. and Rhizobacteria (Ex. *Bacillus* spp., *Paenibacillus* spp. and fluorescent Pseudomonads).
- Specific non-pathogenic strains of *Fusarium* spp., isolated from wilt-suppressive soils, were the most effective antagonists for the reduction (50-80%) of *Fusarium* wilt diseases of tomatoes, watermelons and muskmelons

Core or Spongy Rot - *Botrytis gladiolorum*

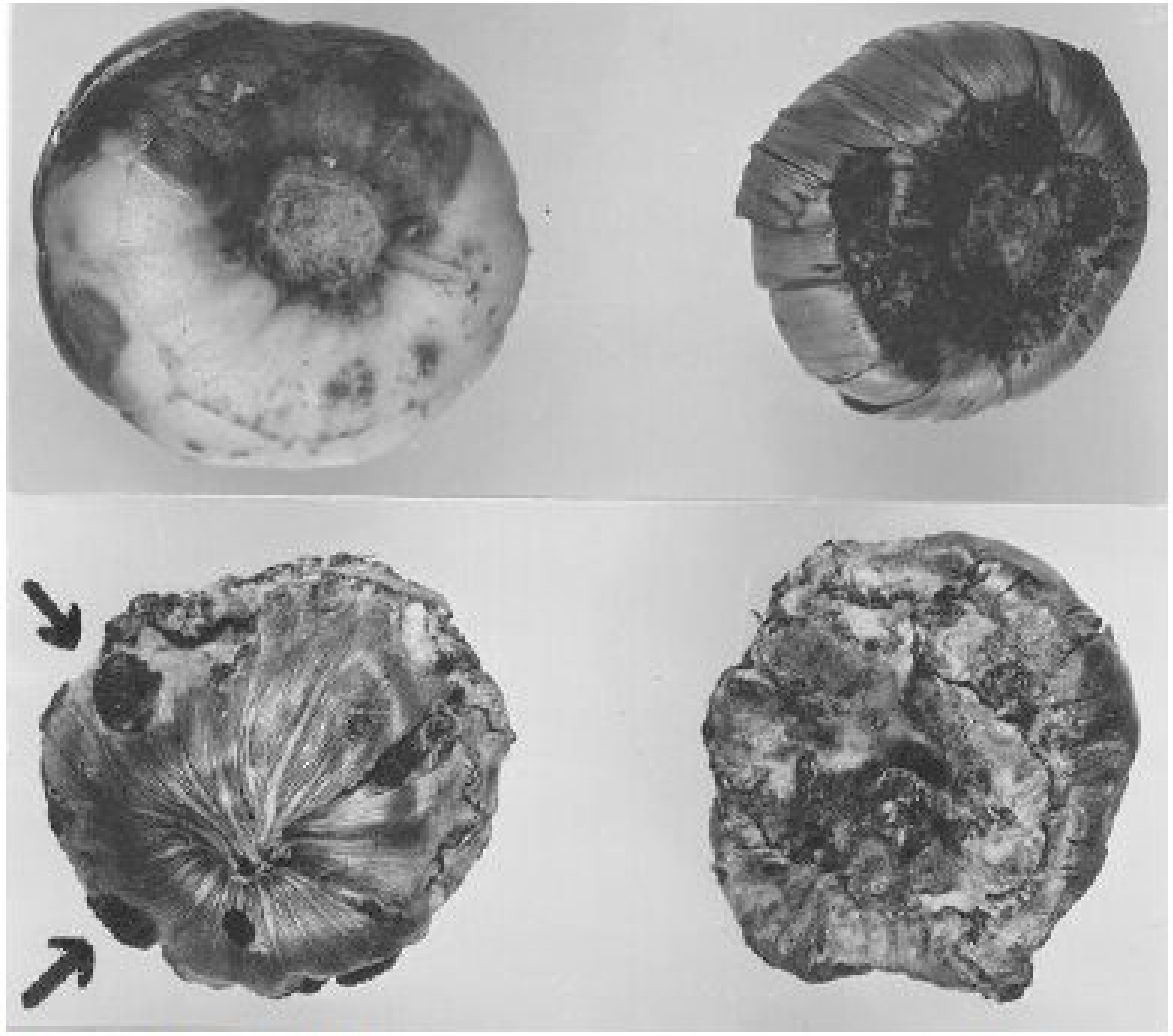


Figure 3. Gladiolus corms in various stages of Botrytis rot. The husks have been removed from the corm at the upper left to show the brown, decayed tissue. Black sclerotia (arrows) may be seen protruding through the husks of the other corms.

Core or Spongy Rot - *Botrytis gladiolorum*

Symptoms:

- Several types of symptoms, **slightly sunken, round, straw coloured or dark greenish brown to dark brown spots** on the corm surface
- These lesions often appear water soaked and vary in size from **pin point to half inch in dia**
- Occasionally the **basal plate turns brown**
- **One or more brown vascular bundles** may extend upto the stem area
- **The core may partially or completely rotted**, with or without decaying of surrounding tissue and with or without brown vascular strands radiating to the surface
- The core may drop away, leaving a doughnut like appearance or completely covered

Core or Spongy Rot - *Botrytis gladiolorum*

Symptoms contd….

- In other instances, **the decay is general, varying in texture from soft and spongy to rather fibrous and firm**
- The **husks may appear normal** but the corm beneath is completely rotted
- In early stage, **water can be squeezed from such affected corms,** in advanced stages the decay becomes **spongy or corky, tan to chocolate brown**, and corms is **usually very light in weight**
- **When the husks are removed**, either **a dark punky mass** interspersed with whitish pockets of mold (mycelium of Botrytis) or it is **rock hard and mummified**.
- Flat **oval to irregular black sclerotia**, 1/8 to ¼ inch long , **often form on the corms in storage or are intermingled with the rotted tissue** and mycelium of the fungus

Core or Spongy Rot - *Botrytis gladiolorum*

Symptoms contd…

- **Infected corms give rise to weak, yellow shoots** which will die soon, while a few only produce healthy shoots
- The fungus produces **spots on leaves and flowers and rotting of stem and leaf bases**
- **Infected parts covered with grey mass of spores**

Core or Spongy Rot - *Botrytis gladiolorum*

Management

- It is a soil borne highly destructive fungus which affects the plants during **cool and humid weather**, most favorable temperature being 13 to 15°C.
- The disease can be controlled effectively by spraying fungicides such as Vinclozolin.
- Mancozeb sprays @ 0.2% also control.
- **Hot water treatment (52°C) of corms** is effective in eradicating the pathogen.
- **Proper drying of corms** before storage.

Dry or Neck Rot - *Stromatinia gladioli*

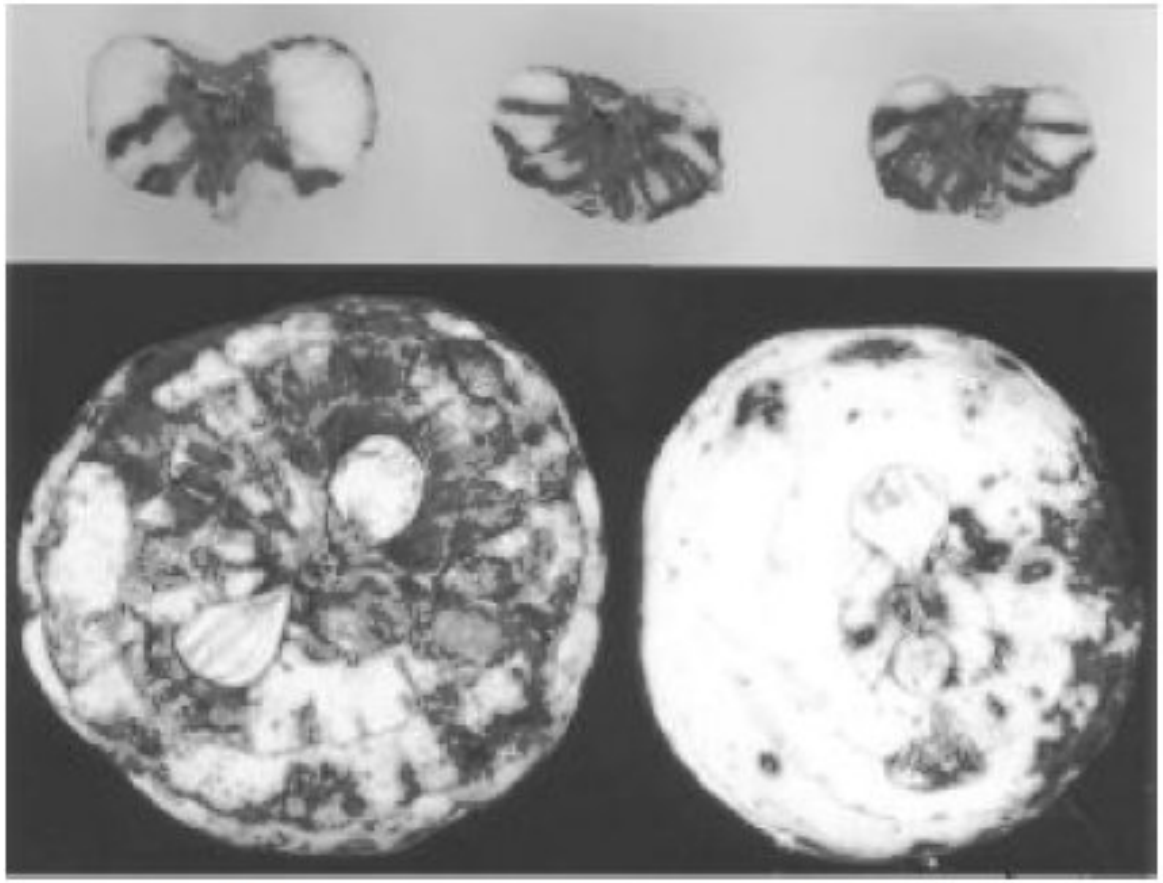


Figure 5. Stromatinia dry rot. Top: Blackened vascular strands extend through the flesh of infected corms. Bottom: Stromatinia lesions on gladiolus corms. All of the husks except the bud scales ahve been removed to show the lesions in the flesh of the corms (IL Natural History Survey photo).

Dry or Neck Rot - *Stromatinia gladioli*

Stromatinia gladioli Produces dry rot of all below ground plant parts

Symptoms

- **Young corms tend to be more susceptible** than older ones.
- **On the husks -** tobacco brown lesions
- **Diseased corms are characteristically have minute and reddish brown lesions** ranging in size from pin points to about 1 ½ inch dia
- The lesions are at first usually develop **on the side and lower half of the corm**
- The spots frequently appear on the upper half as well
- As **lesions enlarge, the centres become sunken with** definite **slightly raised margins** and usually turn black.
- The lesions often merge into large irregular areas

Dry or Neck Rot - *Stromatinia gladioli*

Symptoms contd..

- Very small **sclerotia form on the husks, in corm lesions and on dead stem**
- When infected corms are cut vertically in half, blackened **vascular strands** can be seen that extend from the **core to the surface of the corm**
- **The decayed tissue is corky in texture and mummification of the corms often occurs in storage**

Dry or Neck Rot - *Stromatinia gladioli*

Symptoms

- The fungus enters leaves **at or near the ground line** and causes **a dry rot of leaves and stem,** which may kill the plant.
- Rotting of leaf bases often results in premature yellowing and death of the tops
- The **line separating diseased and healthy is rather sharp**
- The **neck-rot phase of the disease** occurs primarily in early summer.
- The leaves turn brown from the tip downward and decay at the base causing **neck rot but corms remain attached to the stem firmly**.

Dry or Neck Rot - *Stromatinia gladioli*

Pathogen

- A soil borne fungus, the infection **originates from infected corms and sclerotia**, survives in soil indefinitely (> 10 years).
- Sclerotia germinate in response to nearby gladiolus roots.
- The fungus produces sclerotia frequently **but apothecial formation is not common in nature.**
- High proportion of buried sclerotia in soil increase spread of disease by close planting

Host range: Crocosmia, crocus, and freesia.

- The disease is **favored by cool, wet conditions, and wet soils.**
- Often, secondary organisms enter the infected stem at ground level, causing a wet or slimy appearance.

Storage Rot - *Penicillium gladioli*

Symptoms

- Diseased corms **show numerous round black and small lesions varying in size** with slightly raised edges which may coalesce together forming irregular area.
- The disease mostly appears in the form **of black, brown, greenish or yellowish mouldy growth on the corms during storage**.
- **Under poor air circulation the corms may** rot and emit a foul smell.
- As infection occurs through injuries, **wounding of corms should be avoided** at the time of digging or during handling.

Storage Rot - *Penicillium gladioli*

Management

- Four to six years crop rotation is essential to remove the soil borne inoculums.
- Hot water treatment is effective in eliminating the pathogen.
- **The disease can be managed** by soaking the cormels in cool water for 24 hours followed by hot water at 54.5^0C/30 minutes
- **Corms should be treated** with Thiram (0.3%).
- The corm after digging out from the soil should be treated with Thiram 75 DS (0.3%) and properly cured **at 30-35^0 C for 7-10 days** before they are taken to the cold store.
- Damp storage conditions should be avoided.
- High temperature (more than 50^0 C in the cold store) also needs to be avoided, as it could lead to rapid rotting of corms under humid conditions

Anthracnose - *Colletotrichum gloeosporioides*

Aster phyllody - Phytoplasma

Aster phyllody - Phytoplasma

- Phyllody disease on aster was first reported during 1988 from Bengaluru, India
- Chlorosis
- Upright growth
- Small leaf
- Short internode
- Stunting
- Profuse vegetative growth - **Compact bunch like**
- Phylloid flower (**transformation of floral organs into leaf-like structures**)
- Known to be transmitted by leaf hopper, *Orosius albicinctus*

Aster yellows - Phytoplasma

Vein clearing on China aster.

Aster yellows - Phytoplasma

The symptoms of aster yellows varies from affecting only part of the flower (L), to changing the color of the petals (C), to causing abnormal growth (R).

Aster yellows - Phytoplasma

Virescence caused by aster yellows.

Phyllody caused by aster yellows.

Aster yellows - Phytoplasma

Symptoms

- Plants infected when small tend to be stunted, may develop rosetting of leaves, and affected leaves are often narrower than healthy leaves.
- Top leaves may be chlorotic (yellowed) erect, may have vein clearing, or may develop **reddish, bronze or purple tinges.**
- The most dramatic symptoms occur on reproductive parts of the plant **(sterile)**
- The flowers are frequently deformed, with **underdeveloped petals or weird, extra growths, or a change of color.**
- Flower heads open **unevenly or only partly** and have a yellowish-green cast.
- Plants with phyllody have tufts of pale, deformed leaves, often inside the flower or in place of the petals.

Aster yellows - Phytoplasma

Transmission

- The disease is spread by the **six-spotted leaf hopper -*Macrosteles divisus, M. quadrilineatus***

Host Range

- It affects calendula, coreopsis, cosmos, delphinium, daisy, marigold, petunias, phlox, zinnia, and many other flowers and weeds.
- Carrots, celery and lettuce are also attacked.

Aster yellows - Phytoplasma

Management

- Prompt removal and destruction of affected plants when noticed.
- Control weeds such as dandelions and plantain, which may harbor
- The use of oat straw mulch may reduce leaf hopper numbers.
- Spray insecticides metasystox or monocrotophos or acephate or imidacloprid to control leaf hopper